WEN-HORNG SHEU

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Education

Ph.D. in Computer Science, University of California, Davis; GPA: 4.0/4.0 *2023-present*
- Research Areas: Distributed Algorithms, Streaming Algorithms, Large-Scale Computation.

M.S. in Computer Science, National Tsing Hua University; GPA: 3.9/4.0 *2019-2021*
- Coursework: Approximation Algorithms, Computational Geometry.

B.S. in Computer Science, National Tsing Hua University; GPA: 3.85/4.0 *2015-2019*
- Coursework: Introduction to Machine Learning, Parallel Algorithm Design.

Selected Publications

Fixed-Threshold Peeling in Sublinear MPC: Round-Approximation Tradeoffs and Applications, DISC '26
- Designed distributed algorithms for k-core and densest subgraph, with applications in community detection.
- Reduced the round complexity from $O(\log^{1/2} n)$ to $O(\log^{1/4} n)$ via a pruned graph exploration technique.

Faster Semi-streaming Matchings via Alternating Trees, ICALP '25
- Designed a streaming algorithm for computing a $(1+\epsilon)$-approximate maximum matching.
- Improved a STOC '22 result, reducing the pass complexity from $O(\epsilon^{-19})$ to $O(\epsilon^{-6})$ via a new data structure.

A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic, SPAA '25
- Designed a framework for $(1+\epsilon)$-approximate matching in distributed and dynamic settings.
- In distributed settings, reduced the round-complexity dependence on ϵ from $\Omega(\epsilon^{-52})$ to $O(\epsilon^{-7}\log(1/\epsilon))$.

A Kernelization Algorithm for Finding a Perfect Phylogeny from Mixed Tumor Samples, TCBB '25
- Designed a preprocessing algorithm for inferring evolutionary relationships from mixed tumor samples.
- Accelerated existing algorithms when the degree of mixing within tumor samples is bounded.

Software Project

Critical Path Reporting for Static Timing Analysis (Second Place in ACM TAU 2018 Contest)
- Contributed to a C++ tool of 5,000+ lines of code for analyzing digital circuits specified in Verilog format.
- Implemented Eppstein's k-shortest path algorithm for enumerating critical paths through given logic gates.

Selected Honors

Shadow Program Committee Member for ACM SPAA 2026
- Invited to serve on the shadow program committee for a leading conference in parallel algorithms.
- Reviewed 6 submissions; participated in committee discussions and contributed to paper evaluations.

Gold Award in the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest
- Competed in 6+ international programming contests, including multiple ICPC regional contests.

Grandmaster on Codeforces
- Ranked as a Grandmaster, within the top 1% globally on Codeforces, a competitive programming platform.
- Placed in the top 1% among 10,000+ global contestants in four Codeforces programming contests.

Skills

Programming Languages	C++, Python
Tools	Git, LaTeX, LiteLLM, Linux/Bash
Languages	English (fluent, TOEFL 105/120), Chinese (native)
Coursera Certificate	AI Agents and Agentic AI with Python & Generative AI